

Appendix B

Hypothesis Testing

One-Sample Hypothesis Test

Type	Sample	Population	σ	Test Statistic
mean	$n \geq 30$	any	known	$z_{\bar{x}} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
mean	$n \geq 30$	any	unknown	$z_{\bar{x}} = \frac{\bar{x} - \mu}{s/\sqrt{n}}$
mean	$n < 30$	normal	known	$z_{\bar{x}} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
mean	$n < 30$	normal	unknown	$t_{\bar{x}} = \frac{\bar{x} - \mu}{s/\sqrt{n}}$
proportion	$np \geq 5$ $nq \geq 5$	any		$z_p = \frac{\bar{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$

corresponding

$\bar{p}$ (sample)

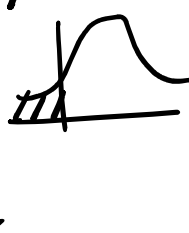
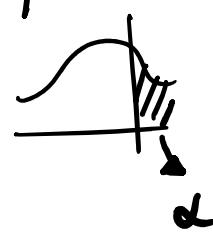
p (population)

corresponding

corresponding

H_0 always contains "="
 $\mu = c \quad \mu \leq c \quad \mu \geq c$

H_1 never contains "="
 $\mu \neq c \quad \mu > c \quad \mu < c$



$$s = \sqrt{\frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}}$$

critical

$$\chi_c = \mu + z_c \left(\frac{s}{\sqrt{n}} \right)$$

σ_s